

Serving Engine Integration

OpenAI-compatible
API gateway

Continuous-batching
decode engine

Session Manager
(residency & lifecycle)

MLXDKVWrapper — generate(): chunked prefill + decode loop

MLXQwenModel — native KV cache (prefill only) + prefill→decode release

Patched Qwen2 Attention

prefill (L > 1)

decode (L = 1)

Prefill Path

Block-sparse causal attention
over the native cache

Capture K/V → Stream-compress
blocks leaving the window

write

Decode Path

Ingest token into dense window
(+ flush eligible block)

**Route top-K → Fused low-rank +
exact-residual + recency attention**

read (no decompress)

MLXKVBlockManager — per-session KV store (×28 layers)
dense recency window $[H_{kv}, 1,280, d]$ fp16 (newest 1,024 tokens)
+ compressed pool $\{U, V_K, V_V, \text{anchors, residuals, min/max}\} \times 256$ blocks